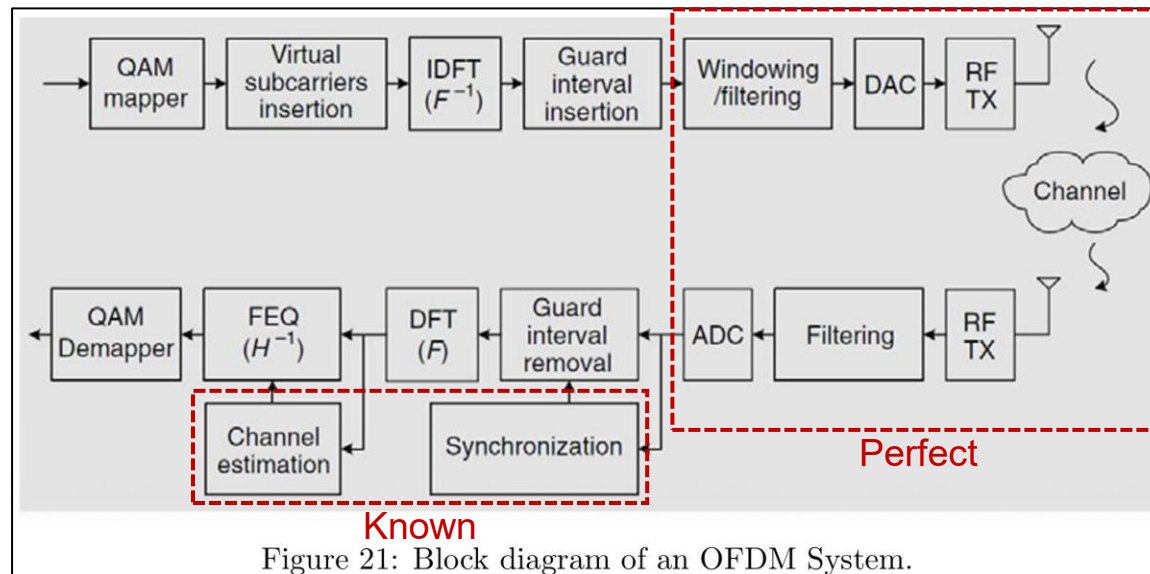


■ Demonstration1 (OFDM):

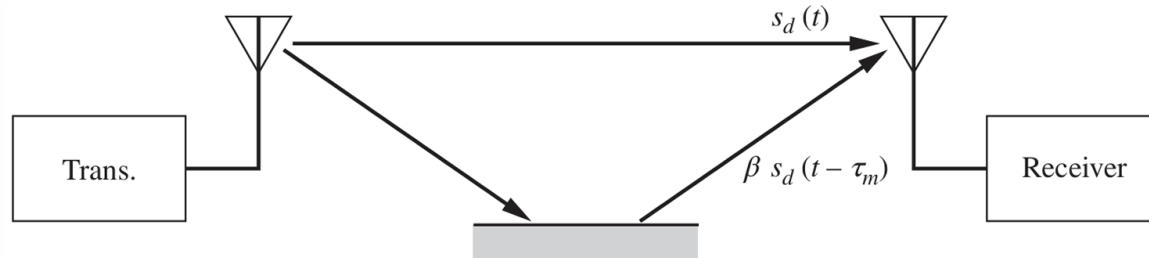
- Construct the transmitter of an OFDM system.
- Observe the spectrum of the signal.
- Parameters (for reference)
 - Number of OFDM symbols (=2400)
 - FFT size (=128)
 - CP size (=FFT size*1/8)
 - Number of active subcarriers (=24)



■ Homework1 (OFDM):

- Consider only two-ray multipath (without noise):
$$0.9 s_d(t) + 0.6s_d(t - 4)$$
- Construct the receiver of OFDM and implement one-tap equalizer, plot real part and calculate MSE.
- Adjust the second path delay to find out the maximum delay that MSE increase rapidly.

■ Consider a two-ray multipath model:



■ The received signal:

$$y(t) = s_d(t) + \beta s_d(t - \tau_m)$$